From Optimal Control to Diffusion Policy

A Trajectory-Optimization $\rightarrow$ TVLQR $\rightarrow$ Behavioural-Cloning Pipeline for Double-Pendulum Swing-Up

Group Project TN2 — Technical Documentation

June 26, 2026

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1 Introduction

The goal of this project is to compare the performance of 2 different learning- based control methods. The first method is a behavioural cloning approach, where we use a dataset of expert demonstrations to train a policy that mimics the expert's actions. The second method is a diffusion policy approach, where we use a diffusion model to learn a policy that can generate actions based on the current state of the system. We will evaluate the performance of both methods in the **double pendulum swingup task** and the **stacking problem**.

1.1 Behavioural Cloning

1.2 Diffusion

In order to understand the diffusion policy approach, we first need to understand the concept of diffusion models. Diffusion models are a class of generative models that learn to generate structured data (in our case, action sequences) by iteratively denoising a sample of noise.

Parameter Setup

For the diffusion model, we first need to define the time parameters of the noising and denoising process.

- **Time Horizon** T is the number of time steps in the diffusion process.
- β **Schedule** β_t (with t $\in \{1, 2, \ldots, T\}$) is a set of hyperparameters that control how quickly the noise is added to the data during the forward diffusion process. It is a monotonic increasing sequence of values that determine the variance of the noise
- **Noise Schedule** $\alpha_t = 1 - \beta_t$ is a set of hyperparameters that control how quickly the noise is removed.
- **Cumulative Noise Schedule** $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$

So we can define the forward diffusion process as follows:

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \tag{1}$$

where x_0 is the original data (action sequence) and x_t is the noisy version of the data at time step t. So now we need to define a procedure to undo the noise in T steps. The reverse diffusion process is defined as follows:

$$x_0 = \frac{1}{\sqrt{\bar{\alpha}_t}} \cdot x_t - \frac{\sqrt{1 - \bar{\alpha}_t}}{\sqrt{\bar{\alpha}_t}} \cdot \epsilon_\theta(x_t, t), \quad \epsilon_\theta(x_t, t) \sim \mathcal{N}(0, I) \tag{2}$$